

Table S1. Descriptive ordinary least-squares fits of the linearized apparent pseudo-first-order representation, $\ln(C_0/C_t) = b + k_{\text{app}}t$, over the complete 0-120 min interval.

Nominal Cu-loading (%)	k_{app} (min ⁻¹)	Intercept, b	R^2
0	0.03588	0.4169	0.9686
3	0.00754	0.5864	0.6830
5	0.03756	0.7692	0.9310
7.5	0.03895	0.0764	0.9662
10	0.01923	1.0244	0.8169

Note: The regressions were fitted separately for each composition using all 13 time points. The intercept was estimated rather than constrained to zero. The reported R^2 values are in-sample goodness-of-fit statistics. Because C_0 represents the post-dark baseline and independent dark-only and catalyst-free photolysis controls were unavailable, the fitted k_{app} values are descriptive apparent removal-rate parameters and should not be interpreted as intrinsic photocatalytic rate constants.